

Syllabus: Text as Data

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https://github.com/PeterTolochko/CEU_TAD_2026

Course Description

This course introduces the theory and practice of using text as quantitative data in social science. We will move from research questions and corpus construction to word counts, classification, topic models, embeddings, and language models. Throughout, we will ask what a representation captures, what it leaves out, and how to evaluate the claims made from it. Classes combine discussion, worked examples, and programming exercises.

Learning Objectives

By the end of the course, students will be able to:

- Formulate a research question that can be investigated with textual evidence and select an appropriate corpus and unit of analysis.
- Build text representations and explain the consequences of preprocessing and feature choices.
- Apply dictionary methods, supervised classifiers, and unsupervised models to textual data.
- Use embeddings and language models for text analysis and assess their limitations.
- Evaluate text measures and model outputs against human judgments and relevant evidence.
- Produce a reproducible analysis and critically discuss validity, bias, and ethical issues.

Tentative Weekly Schedule

Validation, bias, privacy, and reproducibility are addressed throughout the course.

Week 1: Introduction to text as data and text representation

Research questions, corpus and unit of analysis; document-term matrices, vector spaces, and similarity.

Week 2: Preprocessing and feature engineering

Tokenization, stemming and lemmatization, stopwords, n-grams, TF-IDF, and sensitivity to representation choices.

Week 3: From concepts to labels: dictionaries and annotation

Regular expressions, dictionary methods, coding instructions, human annotation, and validation of labels.

Week 4: Supervised text classification

Naive Bayes and linear classifiers; train/test splits, precision, recall, and systematic error analysis.

Week 5: Unsupervised learning and topic models

Clustering and topic models; interpreting topics through words and documents; model selection and stability.

Week 6: Text scaling and substantive interpretation

Measuring positions from text; assumptions of scaling models; validation using metadata and external evidence.

Week 7: Language models

Probabilities of word sequences, n-gram models, smoothing, next-word prediction, and the transition to neural language models.

Week 8: Word embeddings and semantic measurement

Word2vec and GloVe; semantic similarity, semantic axes, and changes in word meaning.

Week 9: Transformers and contextual embeddings

Attention, contextual representations, BERT-style and GPT-style models, and document embeddings.

Week 10: Large language models and automated annotation

Zero-shot and few-shot prompting, codebooks, structured outputs, and validation against human labels.

Week 11: Fine-tuning and model adaptation

Pretraining versus fine-tuning; adapting classifiers, comparing with prompting, domain shift, and robustness.

Week 12: Research design and causal inference with text

Text as treatment, outcome, or control; measurement error; project designs and validation plans.

Software and Prerequisites

Working knowledge of Python or R and basic familiarity with machine learning are expected. Class examples and notebooks use Python; students may use Python or R for assignments.

Assessment

- Homework assignments: 40%
- Final project: 50%
- Participation: 10%

Assignment dates, submission arrangements, and the final-project deadline will be announced separately.

Final Project

Apply one or more text-analysis methods to a substantive or methodological research question. Motivate the question, explain the corpus and method, present results, evaluate the analysis, and discuss limitations. The proposed format follows last year: approximately 6-8 pages of main text, excluding references, figures, and appendices, accompanied by runnable, commented code. Detailed instructions will be provided separately.

Readings

There are no mandatory textbook readings. Recommended references include:

- Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). Text as Data: A New Framework for Machine Learning and the Social Sciences. Princeton University Press. Research design, measurement, and social-science applications.
- Jurafsky, D., & Martin, J. H. Speech and Language Processing, third-edition online draft. Natural language processing, embeddings, and language models. <https://web.stanford.edu/~jurafsky/slp3/>
- James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). An Introduction to Statistical Learning: with Applications in Python. Springer. Model fitting and evaluation. <https://www.statlearning.com/>
- Manning, C. D., Raghavan, P., & Schütze, H. (2008). Introduction to Information Retrieval. Cambridge University Press. Tokenization, TF-IDF, and vector-space methods. <https://nlp.stanford.edu/IR-book/>

Selected research papers and documentation will accompany class topics. Slides, notebooks, and teaching data are available through the class repository.

Participation

Students are expected to prepare the required software, take part in discussions and exercises, and explain their analytical choices. Questions about confusing examples, unexpected outputs, and alternative interpretations are part of the work of the course.

Academic Integrity and Use of Generative AI

Submit work that you can explain and take responsibility for. Cite external sources, datasets, code, and models, and acknowledge assistance. Generative AI tools may support learning, but their use must be disclosed, for example in code comments or a short methods note. State what the tool was used for and check its outputs. You remain responsible for the correctness, originality, and integrity of the submission.